Trio or singleton testing? A published case: singleton exome + parental testing proves a de novo variant

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Pan J, Zhang Y, Hou J, Shi N, Qu H, Jiang L, Liu H. *Case Report: Whole-exome sequencing revealed a de novo variant in SETBP1 gene in a Chinese family with developmental delay.* Frontiers in Genetics. 2025;16 (published 7 November 2025). Department of Reproductive Medicine, 960th Hospital of the PLA Joint Logistics Support Force, Jinan, China. Open access (CC BY). frontiersin.org · 10.3389/fgene.2025.1637931

The patient

- Girl, 5 years; eldest child of healthy, non-consanguineous parents
- Term birth, birth weight 3,200 g
- Motor and language delay, limited spontaneous speech, poor coordination, attention problems
- Full-scale IQ 87 (Wechsler); no seizures

Work-up

- Brain MRI: normal
- EMG: normal
- Metabolic screening: normal
- **Singleton** exome sequencing of the child (trio exome was not done)

Result and how parental testing helped

Exome found a heterozygous nonsense variant **SETBP1 c.1630C>T, p.(Arg544Ter)** (NM_015559.3). Targeted **Sanger sequencing of both parents and the unaffected younger sister** showed the variant was absent in all three, so it arose **de novo**. Confirmed de novo status supported classification as pathogenic (ACMG/AMP) and the diagnosis of **autosomal dominant intellectual disability 29 (MRD29, SETBP1-related)**. The family received genetic counselling for future reproductive planning. The article does not report treatment or follow-up outcome.

Talking points for the doctor visit

- Parents' DNA turns a candidate variant into a confident diagnosis: confirmed de novo is strong evidence (PS2).
- Two routes give this answer: **trio exome** from the start (faster, fewer variants to review) or **singleton exome + targeted parental Sanger** (lower upfront cost).
- Prefer trio for critically ill newborns, severe developmental delay or multiple anomalies, and prenatal anomalies.
- Collect parents' EDTA samples with the child's sample when possible, with full clinical details.
- For a de novo variant, recurrence risk is usually low but not zero (parental germline mosaicism); counsel the family.

For healthcare professionals' general awareness. Not a substitute for clinical judgement; facts and guidelines may change. Case details are as published by the original authors; consult the full article.

EPISODE 3 · QUICK GUIDE

Trio vs singleton: what parents' samples add

Term / reason	What it means
Singleton / duo / trio	Patient only / patient + one parent / patient + both parents. Applies to exome or genome sequencing.
De novo	Variant present in the patient and absent in both parents. Confirmed de novo = strong ACMG evidence (PS2); a common cause of severe neurodevelopmental disorders.
Phase (recessive)	Two variants in one gene: in trans (one from each parent) = compound heterozygous; in cis (same copy) usually does not explain a recessive disease. Parents' data show the phase.
Filtering	Parents' data reduce the number of variants to curate; reports are faster and often have fewer VUS.
Cautions	A variant in a healthy parent is not always benign (reduced penetrance, variable expressivity, parental mosaicism). Consent should cover possible unexpected family-relationship findings.

Study	Finding
Tan et al., Eur J Hum Genet 2019 (Australia), 30 children, both analyses	Singleton exome 36.7% vs trio 40.0%. Trio removed the need for parental Sanger testing but nearly doubled total cost; singleton first with reflex to trio was cost-saving.
Kaschta et al., Kiel (Germany), 448 patients, genome sequencing	Prospective yield: trio GS 36.8%, singleton GS 30.4%, standard exome 36.0%.
Bajaj et al., J Hum Genet 2025 (Mumbai), 137 children with autism, trio exome	16.1% P/LP (22/137); syndromic 35% vs non-syndromic 8.2%; 91.3% of significant variants de novo.

References

- Pan J et al. Front Genet 2025;16. 10.3389/fgene.2025.1637931
- Tan TY et al. A head-to-head evaluation of the diagnostic efficacy and costs of trio versus singleton exome sequencing analysis. Eur J Hum Genet 2019;27(12):1791–1799. record
- Kaschta D et al. Evaluating genome sequencing strategies: trio, singleton, and standard testing in rare disease diagnosis. PMC12445032
- Bajaj S et al. Trio exome sequencing in 137 Indian children with autism spectrum disorder. J Hum Genet 2025;70:611–624. nature.com/articles/s10038-025-01368-4

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